

HAM: A Training-Free Style Transfer Approach via Heterogeneous Attention Modulation for Diffusion Models

Yeqi He^{1,2*} Liang Li^{2†} Zhiwen Yang^{1,2*} Xichun Sheng³ Zhidong Zhao¹ Chenggang Yan¹

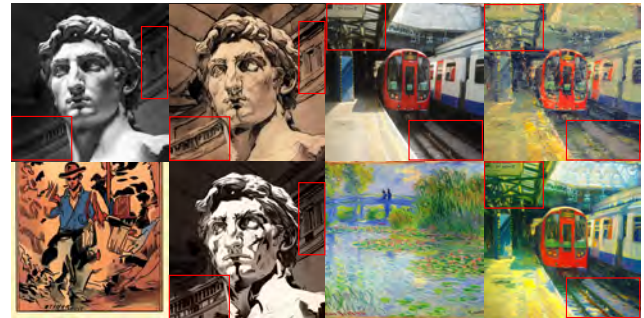
¹ Hangzhou Dianzi University ² Institute of Computing Technology, Chinese Academy of Sciences ³ Macao Polytechnic University
{yeqihe, zhiwen.yang, zhaozd, cgyan}@hdu.edu.cn liang.li@ict.ac.cn p2314922@mpu.edu.mo

Abstract

Diffusion models have demonstrated remarkable performance in image generation, particularly within the domain of style transfer. Prevailing style transfer approaches typically leverage pre-trained diffusion models' robust feature extraction capabilities alongside external modular control pathways to explicitly impose style guidance signals. However, these methods often fail to capture complex style reference or retain the identity of user-provided content images, thus falling into the trap of style-content balance. Thus, we propose a training-free style transfer approach via **heterogeneous attention modulation (HAM)** to protect identity information during image/text-guided style reference transfer, thereby addressing the style-content trade-off challenge. Specifically, we first introduces style noise initialization to initialize latent noise for diffusion. Then, during the diffusion process, it innovatively employs HAM for different attention mechanisms, including Global Attention Regulation (GAR) and Local Attention Transplantation (LAT), which better preserving the details of the content image while capturing complex style references. Our approach is validated through a series of qualitative and quantitative experiments, achieving state-of-the-art performance on multiple quantitative metrics.

1. Introduction

The development of generative diffusion models has propelled advances in text-to-image generation [1, 7, 25, 27–29], image editing [15, 20, 35], and related fields. Among these, the generative diffusion models have also been applied to style transfer [13, 43], specifically migrating the style references of a given content image to a designated style preset while preserving its identity information. Given the powerful text-to-image generation capabilities of gen-



(a) Image-guided style transfer results. Compare with StyleID [3].



(b) Text-guided style transfer results. Compare with DiffArtist [14].

Figure 1. Comparative results of style transfer methods: Content image (top-left), style reference (bottom-left), baseline method (top-right), and our HAM (bottom-right). Red boxes denote significant identity retention disparities.

erative diffusion models, they have brought a efficient yet challenging new paradigm to style transfer.

Diffusion-based style transfer approaches typically utilize the inherent generative capabilities of pre-trained diffusion models [7, 25, 29] to achieve style migration for given content images. Some approaches, involves explicit style-content feature decoupling [9, 17, 37, 39], leveraging interpretability to process style features while retaining content features, and incorporates style attributes into pre-trained diffusion models via LoRA [12] or ControlNet [41] for fine-tuning-based style control. While effective for style transfer, these methods are computationally intensive and lack

*This work is done during the intern in VIPL group, ICT, CAS.

†Corresponding author

robustness, as their performance on diverse style references is highly sensitive to the extent of fine-tuning.

To fully utilize the powerful generative capabilities of diffusion models, training-free style transfer methods have been proposed. Representative works like StyleID [3] and DiffArtist [14] achieve stylization by injecting the keys and values from style features, extracted via diffusion model inversion, into the self-attention layers during generation. As theory indicates [10, 30], self-attention features, including queries, keys, and values, collectively encode various semantic and spatial relationships. Consequently, such methods that solely rely on self-attention manipulation results in insufficient style or distorted content, as Fig. 1 demonstrates, resulting in an imbalance between style and content.

In this work, we propose a training-free style transfer approach via **heterogeneous attention modulation** for diffusion models (**HAM**), which significantly improves the style-content balance capability of style transfer. Our method utilizes a style/content teacher model obtained from style references and content images, and then uses HAM to combine and share the knowledge from the teacher model to the student generator, thereby achieving style transfer.

Building upon framework process, we propose a style-infused **noise initialization (SINI)** at timestep T , where the initial latent noise is derived by fusing the inverted noise from the reference style and content images through adaptive instance normalization. To better preserve identity information, the fused initial latent noise is then modulated by the inverted content initial latent noise. Subsequently, in the process of generating and diffusing stylized images, we introduce our HAM, comprising the **global attention regulation (GAR)** and **local attention transplantation (LAT)**, respectively. GAR is a mechanism designed to preserve content and introduce style by exerting macroscopic control over attention injection, thereby maintaining the original spatial and style semantic structure. Specifically, it fuses content and style teacher’s attention projections into the specific projections, ensuring their statistical distribution aligns with the corresponding attention projections in the student generator. These specific projections are then reconciled with the ones to further stabilize content/style information. After that, to implement precise style/content control in the absence of text prompts, we introduce LAT through feature operations in cross-modal cross-attention. To better preserve identity information, we inject and weight the query from the content teacher into the query from student generator, while the original key/value pairs are replaced directly with those from the style teacher to ensure effective style guidance. This heterogeneous attention modulation effectively decouples style guidance and content preservation, enabling high-fidelity stylization without compromising structural integrity.

Our GAR and LAT adapt to base model architectures:

operating within self-/cross-attention for SD2.1 [29], and joint-/dual-attention for SD3.5 [7], respectively.

The main contributions are summarized as follows:

- We propose a training-free stylized image generation method, HAM, which can achieve high-quality stylized image generation without the need for gradient optimization of style images.
- In our proposed HAM, GAR effectively macroscopically introduces features from the style/content teacher into the student generator, while LAT precisely controls the guidance between style and content. The combined effect improves the quality of the generated stylized images.
- We demonstrate HAM’s universal compatibility across DDIM-based (SD2.1) and DiT-based (SD3.5) architectures, achieving state-of-the-art performance on multiple metrics through comprehensive evaluations.

2. Related Work

2.1. Text-Driven Image Generation

With the advancement of deep learning [4, 5, 40], Text-driven image generation [1, 27, 28] has enabled the synthesis of highly realistic and semantically coherent images. Advances in text encoder architectures, exemplified by the SD series [7, 25, 29], have significantly improved text-to-image synthesis through structural refinements and systematic optimization. These technical developments yield quantifiable gains in output visual quality, particularly in enhanced texture detail and resolution fidelity, while reinforcing model robustness in maintaining precise semantic alignment with complex, compositional textual prompts. The progress in generative foundations [24, 33] has also propelled developments in related areas, including: (1) text-guided image editing [21, 22, 30], (2) semantic-aware style transfer [13, 43], and (3) emerging multimodal generative applications beyond static imagery [18, 34, 44–47].

2.2. Image Style Transfer

Style transfer aims to apply a reference style’s visual characteristics to a content image while preserving its structural and semantic core, hinging on disentangling content and style representations. Current approaches fall into tuning-based and training-free categories. Tuning-based methods adapt models through parameter updates (e.g., coupling style-specific content or adding lightweight adapters), exemplified by ControlNet [41] training conditional copies, B-LoRA [9] innovating weight optimization, and CSGO [39] using curated adapters. Training-free methods manipulate diffusion mechanisms during inference without parameter changes, altering attention maps/activations to redirect synthesis—e.g., P2P [10] injecting reconstructed cross-attention maps and StyleID [3] fusing features into self-attention layers for zero-shot stylization.

3. Method

In this section, we present our proposed method comprising three core modules: global attention regulation, local attention transplantation, and style-infused noise initialization. Compatible with both DDIM-based SD2.1 [29] and DiT-based SD3.5 [7] architectures, our method is detailed in the following subsections, with technical descriptions primarily based on the SD2.1 framework.

3.1. Preliminaries

Before delving into the specifics of our proposed methodology, we initially provide a comprehensive background on the fundamental techniques that underpin our method.

3.1.1. Latent Diffusion Models

Latent Diffusion Models (LDMs) [29] represent a prominent image generation framework that maps images into the latent space of the Variational Auto-Encoder (VAE) [16] and subsequently leverages the powerful generative capacity of diffusion models to synthesize high-quality images while optimizing computational efficiency. The most representative class of such methods comprises the Stable Diffusion (SD) [7, 25, 29], which utilizes text prompts as input conditions to denoise latent noise for generating high-quality, high-fidelity images. The overall denoising process is formally described by Eq. 1.

$$\mathcal{L}_\theta = \mathbb{E}_{z,t,c,\epsilon \in \mathcal{N}(0,1)} \left[\|\epsilon - \epsilon_\theta(z_t, t, c)\|_2^2 \right], \quad (1)$$

where t denotes the current diffusion time step, z_t represents the latent noise vector corresponding to time step t , c signifies the conditioning text prompt, ϵ is the Gaussian-distributed noise sampled from $\mathcal{N}(0, 1)$, and $\epsilon_\theta(z_t, t, c)$ denotes the noise component predicted by the model.

3.1.2. Attention Mechanism

To enhance the quality of synthesized images and effectively integrate external conditioning information, Stable Diffusion incorporates multiple groups of self-attention and cross-attention blocks [36], which are typically arranged in complementary pairs. As previously established, our proposed methodology focuses on modulating this dual-attention mechanism, specifically by concurrently targeting both self-attention and cross-attention blocks. The precise mathematical formulations governing these two attention operations are formally expressed in Eq. 2.

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V, \quad (2)$$

where Q , K , and V denote the query, key, and value matrices, respectively, with d_k representing the dimensionality of the key vectors. In self-attention, these matrices are all

derived from the same input feature map, enabling intra-modality feature integration. Conversely, in cross-attention, Q is typically projected from the visual feature space, while K and V originate from the conditioning context (e.g., text embeddings), facilitating inter-modality information fusion.

3.2. Global Attention Regulation

As substantiated by prior research [10, 30], self-attention projections within Latent Diffusion Model (LDM)-based generative frameworks [7, 29] inherently encapsulate both spatial positional relationships and semantic representations pertinent to content-related information. Consequently, the preservation of content image identity information during style transfer operations emerges as a critical prerequisite for maintaining structural fidelity. To address this fundamental requirement, we propose a novel feature fusion module that strategically leverages complementary information from both content teacher models and style teacher models, as shown in Fig. 2. This module enables global modulation of the self-attention layers within the main branch’s stylized generation model, thereby ensuring dual objectives: 1) consistent retention of content identity characteristics throughout the stylization process, and 2) effective incorporation of stylistic attributes derived from either image or textual style references. Specifically, the initial phase of our global attention regulation module focuses on synergistic integration of multi-source feature representations. Given the imperative for training-free feature manipulation, we employ adaptive instance normalization [13], a pioneering work in the style transfer domain, to achieve decoupled feature recombination. This operation fuses content-specific attention projections ($Q_{self}^c, K_{self}^c, V_{self}^c$) from content teacher models with style-specific projections ($Q_{self}^s, K_{self}^s, V_{self}^s$) from style teacher models, generating optimized composite projections ($Q_{self}^{cs}, K_{self}^{cs}, V_{self}^{cs}$). The mathematical formalization of this fusion process, which aligns feature distribution statistics while preserving discriminative attributes, is detailed in Eq. 3.

$$\begin{aligned} Q_{self}^{cs} &= \sigma(Q_{self}^s) \cdot \frac{Q_{self}^c - \mu(Q_{self}^c)}{\sigma(Q_{self}^c)} + \mu(Q_{self}^s), \\ K_{self}^{cs} &= \sigma(K_{self}^s) \cdot \frac{K_{self}^c - \mu(K_{self}^c)}{\sigma(K_{self}^c)} + \mu(K_{self}^s), \\ V_{self}^{cs} &= \sigma(V_{self}^s) \cdot \frac{V_{self}^c - \mu(V_{self}^c)}{\sigma(V_{self}^c)} + \mu(V_{self}^s), \end{aligned} \quad (3)$$

where $Q_{self}^c, K_{self}^c, V_{self}^c$ are the content-specific attention projections in the content teacher model, $Q_{self}^s, K_{self}^s, V_{self}^s$ are the style-specific attention projections in the style teacher model, and $\mu(\cdot)$ and $\sigma(\cdot)$ are the mean and variance.

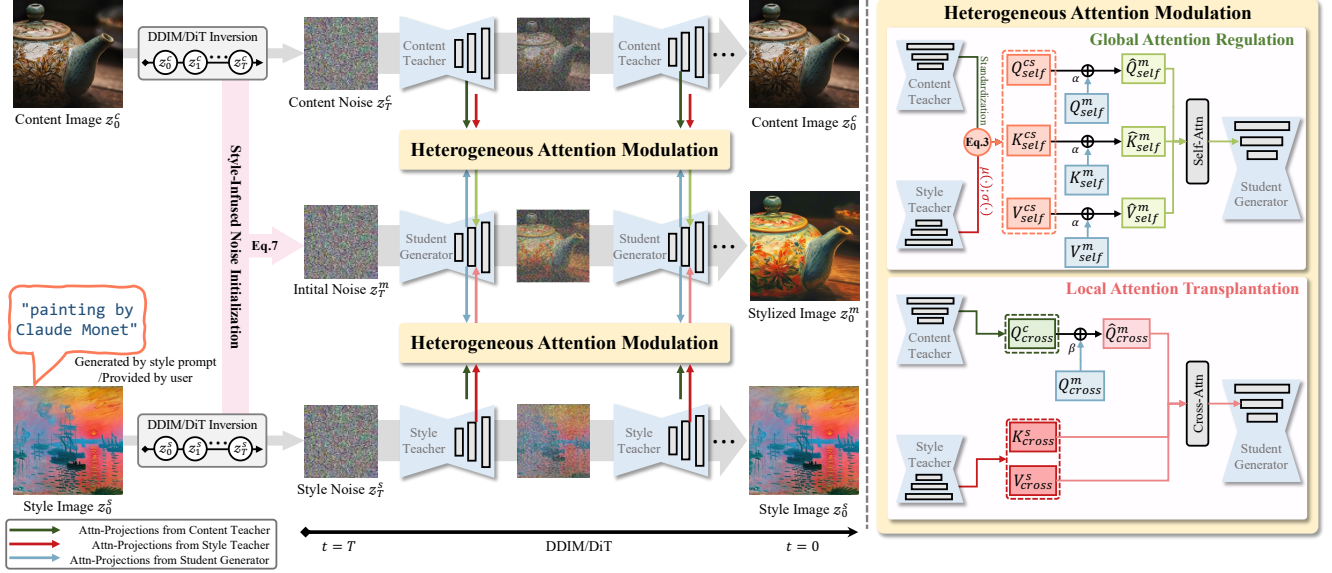


Figure 2. The overall pipeline of our method. Our proposed method consists of three main modules: global attention modulation, local attention transfer, and style injection noise initialization, which act on the self-attention, cross-attention, and noise initialization stages respectively. Through the joint modulation of the three modules, the final stylized image can retain more content identity information and capture and transfer complex style references.

Building upon this foundational alignment, our methodology ensures that the statistical properties of the optimized composite projections $(Q_{self}^{cs}, K_{self}^{cs}, V_{self}^{cs})$ exhibit distributional congruence with the intrinsic self-attention projections $(Q_{self}^m, K_{self}^m, V_{self}^m)$ of the main branch. This critical correspondence establishes the theoretical basis for effective global modulation of the self-attention within the stylized generation model, enabling coordinated feature transformation throughout the denoising trajectory. To implement this modulation, we employ a weighted fusion strategy that integrates the optimized composite projections with the main branch’s native self-attention representations using a predefined hyperparameter. This controlled combination achieves dual objectives: 1) persistent conservation of content identity information throughout stylization, and 2) regulated incorporation of stylistic attributes from style references (style images or textual descriptions). The mathematical formulation in Eq. 4 utilizes a fixed blending coefficient α to explicitly balance the trade-off between content preservation and style infusion, ensuring deterministic transformations independent of optimization dynamics.

$$\begin{aligned}\hat{Q}_{self}^m &= \alpha \cdot Q_{self}^m + (1 - \alpha) \cdot Q_{self}^{cs}, \\ \hat{K}_{self}^m &= \alpha \cdot K_{self}^m + (1 - \alpha) \cdot K_{self}^{cs}, \\ \hat{V}_{self}^m &= \alpha \cdot V_{self}^m + (1 - \alpha) \cdot V_{self}^{cs},\end{aligned}\quad (4)$$

where α is a hyperparameter used to control the weight of the fused attention projections.

Subsequently, the resulting modulated self-attention pro-

jections $(\hat{Q}_{self}^m, \hat{K}_{self}^m, \hat{V}_{self}^m)$ are integrated into the main branch’s stylized generation model’s self-attention, ensuring that throughout the stylized generation process, the self-attention projections maintain content image identity information while simultaneously incorporating style references from style images/text. The final self-attention expression is shown in Eq. 5.

$$\text{Attention}(\hat{Q}_{self}^m, \hat{K}_{self}^m, \hat{V}_{self}^m), \quad (5)$$

3.3. Local Attention Transplantation

Existing methods for attention injection [3, 8] predominantly operate within the self-attention blocks of Stable Diffusion. However, since self-attention projections inherently encode substantial spatial-semantic structures, replacing key and value matrices in these blocks inevitably compromises content identity preservation. Furthermore, modifications to self-attention necessitate additional distribution alignment mechanisms (e.g., attention temperature scaling [3]) to mitigate discrepancies between query representations from different models and their corresponding key/value pairs.

To circumvent these limitations, we propose a novel paradigm shift: leveraging underutilized cross-attention channels for style transplantation. As illustrated in Fig. 2, our local attention transplantation module strategically employs feature representations extracted from both content and style teacher models. While prior methods [3, 10] manipulate attention maps by substituting key/value projec-



Figure 3. Qualitative comparison with existing text-driven and image-driven SOTA methods. For fair evaluation, all methods use fixed random seeds: text-driven methods apply prompts directly, while image-driven methods generate style references via SD2.1 using identical prompts. Our HAM method better preserves content identity while maintaining style transfer semantics.

tions in either cross-attention or self-attention blocks, our method also targets the cross-attention like them. Concretely, we transplant style-specific key and value projections (K_{cross}^s, V_{cross}^s) derived from style teacher models into the main stylization branch, replacing their native counterparts (K_{cross}^m, V_{cross}^m) to achieve localized style injection.

Simultaneously, to prevent content identity degradation during diffusion and counteract potential style intrusion from transplanted projections, we implement a content protection mechanism for query representations. This is achieved through weighted fusion between content teacher model’s query projections Q_{cross}^c and the main branch’s native query projections Q_{cross}^m , ensuring persistent conservation of structural identity. The complete operational formalization is provided in Eq. 6.

$$\hat{Q}_{cross}^m = \beta \cdot Q_{cross}^m + (1 - \beta) \cdot Q_{cross}^c, \quad (6)$$

$$\text{Attention}(\hat{Q}_{cross}^m, K_{cross}^s, V_{cross}^s),$$

where β is a hyperparameter that controls the query projection injection weight of the content teacher model.

3.4. Style-Infused Noise Initialization

As comprehensively delineated in Fig. 2, our framework achieves robust style transfer control throughout the diffusion process via the synergistic operation of global atten-

tion regulation and local attention transplantation modules. The configuration of initial noise consequently emerges as the pivotal remaining determinant of stylization efficacy, given its fundamental role in establishing the generative trajectory’s starting state. While direct transplantation of the content teacher model’s initial noise z_0^c to the main stylization branch represents a conceptually straightforward approach, our quantitative evaluations and qualitative assessments (observed under $\gamma = 1$ settings in Fig. 6 and Tab. 5) demonstrate its inability to achieve meaningful style transfer, primarily attributable to insufficient style integration. Similarly, adaptive instance normalization-based fusion of content noise z_0^c and style noise z_0^s yields composite initial noise z_0^m , yet empirical analysis reveals pronounced content identity degradation (observed under $\gamma = 0$ settings in Fig. 6 and Tab. 5). This phenomenon indicates an inherent optimization conflict between style intensity and content fidelity in conventional fusion paradigms.

To resolve this fundamental limitation, we innovate style-infused noise initialization that incorporates a dedicated content residual noise component atop the baseline AdaIN-fused stylized noise. This dual-component architecture explicitly balances stylization intensity against content preservation, enabling precise calibration of their relative contributions throughout the denoising cascade. The complete mathematical implementation is formalized in Eq. 7.

Method	ArtFID↓	FID↓	LPIPS↓	LPIPS-Gray↓	DINO↑	CLIP-I↑	CLIP-T↑	DC↑	CC↑
DDIM(ICLR'21)	31.149	17.939	0.645	0.554	0.278	0.493	0.192	1.524	1.780
ControlNet(ICCV'23)	24.751	13.472	0.710	0.557	0.513	0.583	0.210	1.831	1.916
StyTR ² (CVPR'22)	17.460	10.433	0.527	0.416	0.433	0.487	0.206	1.729	1.794
InstructPix2Pix(CVPR'22)	28.319	17.657	0.518	0.415	0.575	0.620	0.211	1.908	1.963
InstantStyle(arxiv'24)	27.244	15.249	0.677	0.556	0.474	0.604	0.198	1.765	1.921
CSGO(arxiv'24)	27.116	15.207	0.673	0.527	0.482	0.581	0.197	1.775	1.893
StyleID(CVPR'24)	<u>15.161</u>	8.273	0.635	0.516	0.544	0.619	0.213	1.873	1.964
STAM(CVPR'25)	16.941	9.269	0.650	0.532	0.531	0.608	<u>0.221</u>	1.869	1.963
AttDistillation(CVPR'25)	16.170	<u>8.926</u>	0.629	0.514	0.541	0.615	0.219	1.878	1.969
DiffArtist(MM'25)	16.174	9.641	<u>0.520</u>	<u>0.413</u>	<u>0.629</u>	<u>0.626</u>	0.220	<u>1.987</u>	<u>1.984</u>
HAM(Ours)	15.151	9.244	0.479	0.362	0.728	0.682	0.223	2.113	2.057

Table 1. Quantitative comparison with existing text-driven and image-driven SOTA methods. The best results are highlighted in bold, and the second best results are underlined. The color coding indicates relative performance.

$$\begin{aligned}
z_T^m = & \gamma \cdot \left[\underbrace{z_T^c - \left(\sigma(z_T^s) \cdot \frac{z_T^c - \mu(z_T^c)}{\sigma(z_T^c)} + \mu(z_T^s) \right)}_{\text{Content Residual Noise}} \right] \\
& + \left[\underbrace{\sigma(z_T^s) \cdot \frac{z_T^c - \mu(z_T^c)}{\sigma(z_T^c)} + \mu(z_T^s)}_{\text{Stylized Initial Noise}} \right], \quad (7)
\end{aligned}$$

where γ is a hyper-parameter that controls the weight of the content residual noise, z_T^c is the initial noise of the content teacher model, z_T^s is the initial noise of the style teacher model, z_T^m is the stylized initial noise for the main branch, and $\mu(\cdot)$ and $\sigma(\cdot)$ are the mean and variance.

4. Experiments

4.1. Experiment Setup

Implementation Details We employ diffusion models based on the SD2.1 and SD3.5 architectures. The denoising process utilizes 50 steps for SD2.1. Images are resized to 512×512 pixels, with SD2.1 hyperparameters configured as $\alpha = 0.75$, $\beta = 0.25$, $\gamma = 0.5$. Experiments execute on a single NVIDIA RTX3090, where SD2.1 inversion requires 4s (50 steps) and stylized image generation completes in 16s (50 steps). The results of our method HAM on SD3.5 are discussed in the supplementary material.

Datasets We conduct experiments on the MS-COCO [19] and the WikiArt [32]. Specifically, 1,000 images from MS-COCO were randomly selected as the test content images. For WikiArt, a collection of images from multiple artists was chosen as the style references to represent the style distribution for FID computation. The final test dataset comprises 1,000 content images, 1,000 corresponding generated stylized images, and the corresponding artists' works from WikiArt as style image references.

Comparison Methods We compare with existing text-driven and image-driven methods: DDIM [31], ControlNet [41], StyTR² [6], InstructPix2Pix [2], InstantStyle [37], CSGO [39], StyleID [3], STAM [8], AttDistillation [48] and DiffArtist [14]. For fair text/image-guided comparison, we fix random seeds and use identical prompts: (1) Text-guided: direct prompt input (2) Image-guided: prompts generate style references via SD2.1

Evaluation Metrics We evaluate both traditional metrics FID [11], LPIPS [42], ArtFID [38] and metrics based on DINO [23] and CLIP [26] (DINO, CLIP-I, CLIP-T). For CLIP-T, the input text prompts are style-specific prompts. Our evaluation framework comprises: (1) **Style strength metrics**: FID and CLIP-T measuring stylization degree; (2) **Content preservation metrics**: LPIPS, DINO and CLIP-I assessing content consistency; (3) **Comprehensive style transfer metric**: ArtFID for overall performance. To better comprehensively evaluate stylized images, we introduce two novel composite metrics following ArtFID's computation paradigm: $DC = (DINO + 1) \cdot (CLIP-T + 1)$ and $CC = (CLIP-I + 1) \cdot (CLIP-T + 1)$.

4.2. Performance Evaluation

Qualitative Evaluation As shown in Fig. 3, under our fair evaluation setting, our HAM and existing SOTA methods are compared across diverse text-driven and image-driven style transfer tasks. Due to space constraints, qualitative results omit several methods (e.g., DDIM, ControlNet) that perform poorly in quantitative assessments. Visual results indicate that StyTR² and InstructPix2Pix struggle to capture complex style patterns, leading to noticeable style leakage and content distortion. InstantStyle and CSGO either inadequately represent style details or suffer from severe style-content leakage, resulting in loss of identity. Although StyleID, STAM, AttDistillation, DiffArtist produce reasonable stylizations, they exhibit a consistent

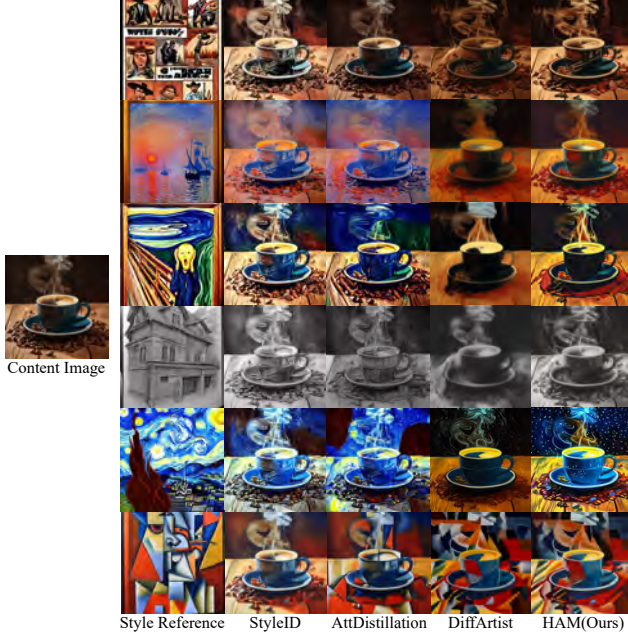


Figure 4. Qualitative results of our method HAM and the SOTA method are presented under different style references for the same content image. It can be observed that our method HAM has significant advantages in both style transfer and identity preservation.

style-content trade-off: either style is well-captured at the cost of content structure, or content is preserved with insufficient style expression. In contrast, HAM accurately captures stylistic attributes under various style-references while maintaining high content fidelity, thereby improving the overall quality of the generated stylized images.

Additionally, as illustrated in Fig. 4, another qualitative experiment is conducted using a single content image under varied style references, with fixed parameters across all methods and no specific adjustments for any styles. The results reveal that existing SOTA methods not only struggle with balancing style and content but also demonstrate limited adaptability when applied to diverse styles using the same content image. In contrast, our HAM shows stronger robustness in transferring complex stylistic information from multiple references to the same content image, while more effectively preserving structural details compared to current SOTA methods. These findings affirm the robustness of our HAM in handling style transfer tasks.

Quantitative Evaluation As shown in Tab. 1, our method HAM achieves optimal (CLIP-T) and near-optimal (FID) performance on two critical style-strength metrics, demonstrating its efficacy in distilling and transferring style reference information. Crucially, the top-ranked CLIP-T performance explicitly confirms HAM’s exceptional alignment with textual style semantics across diverse prompts. For

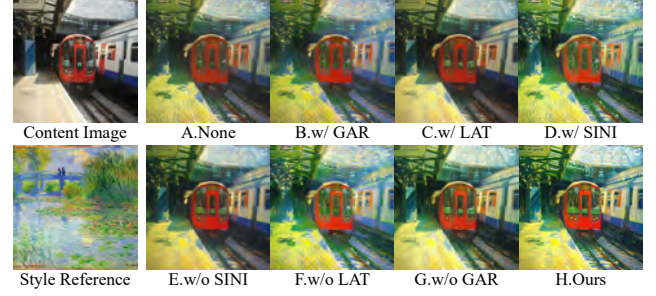


Figure 5. Qualitative ablation study of different modules in our method. The indexes are consistent with those in the quantitative experiments.

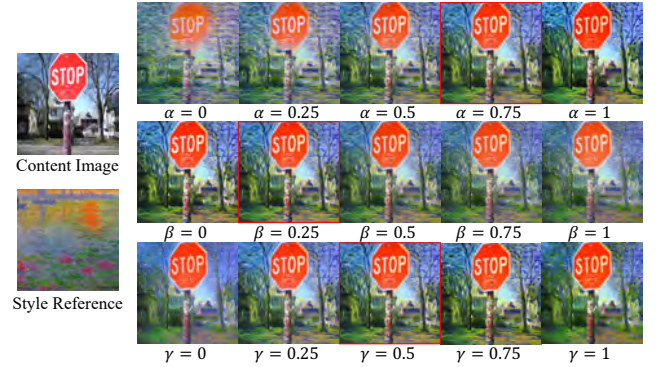


Figure 6. Qualitative ablation study of hyper-parameters in our method. The red boxes represent the hyperparameters we selected.

traditional content preservation metrics (LPIPS and LPIPS-Gray), HAM significantly outperforms all baselines by notable margins, underscoring its absolute advantage in retaining both structural integrity and fine-grained details of content images before and after color removal. This quantitatively reinforces qualitative observations of high content identity preservation. Furthermore, the other content metrics (DINO and CLIP-I) validate HAM’s robustness in preserving inter-image structural coherence, global visual consistency, and visual semantic consistency. This is consistent with our qualitative conclusion regarding generation performance. Finally, regarding overall quality metrics (ArtFID, DC, CC), HAM maintains optimal balance between holistic content preservation and precise style semantics, markedly surpassing all competing methods in synthesizing perceptually harmonious outputs.

4.3. Ablation Study

Ablation on different modules As illustrated in Fig. 5 and Tab. 2, ablation studies evaluate three core modules of the HAM method. Both qualitative and quantitative results demonstrate the contributions of our modules. Global attention regulation module improves the CLIP-T score while slightly increasing DINO and CLIP-I metrics. Qualitative

No.	GAR	LAT	SINI	DINO	CLIP-I	CLIP-T	DC	CC
A	✗	✗	✗	0.609	0.626	0.220	1.963	1.984
B	✓	✗	✗	0.618	0.626	<u>0.231</u>	1.993	2.002
C	✗	✓	✗	0.712	<u>0.696</u>	0.193	2.042	2.023
D	✗	✗	✓	0.588	0.624	0.225	1.945	1.989
E	✓	✓	✗	0.746	0.696	0.202	<u>2.099</u>	<u>2.040</u>
F	✓	✗	✓	0.599	0.627	0.235	1.976	2.010
G	✗	✓	✓	0.695	0.679	0.200	2.034	2.015
H	✓	✓	✓	<u>0.728</u>	0.682	0.223	2.113	2.057

Table 2. Quantitative ablation study of different modules in our method. The indexes are the same as those in the qualitative experiments.

α	DINO↑	CLIP-I↑	CLIP-T↑	DC↑	CC↑
1	<u>0.695</u>	<u>0.679</u>	0.200	2.034	2.015
0.75	0.728	0.682	0.223	2.113	2.057
0.5	0.690	0.667	0.225	<u>2.070</u>	<u>2.042</u>
0.25	0.602	0.634	<u>0.226</u>	1.964	2.003
0	0.497	0.602	0.229	1.840	1.969

Table 3. Quantitative ablation study of hyper-parameter α in Global Attention Regulation (GAR).

results further illustrate its role in enhancing the stylization intensity and content preservation, confirming its consistency with our design objective. This is also consistent with the performance of quantitative indicators. The local attention transplantation module significantly improves DINO and CLIP-I content preservation metrics but adversely affects CLIP-T. This corresponds to qualitatively observed stronger identity retention and reduced stylization, indicating that integrating the style teacher’s key/value requires protecting the main branch’s query with the content teacher’s query features (further evidenced in the next paragraph for the hyperparameter β). For style-infused noise initialization, a modest CLIP-T improvement is observed with no measurable impact on DINO or CLIP-I. Visually, this module enriches stylistic elements and color diversity in the output. Similarly, the results of quantitative and qualitative experiments on SINI are consistent. Collectively, the modules establish HAM’s balance on DC/CC metrics.

Ablation on different hyper-parameters As shown in Fig. 6, ablation experiments assess HAM’s key hyperparameters (α, β, γ) qualitatively. For α in the global attention regulation module, as shown in Tab. 3, content preservation metrics (DINO and CLIP-I) achieve optimal performance at $\alpha = 0.75$, degrading at other values, consistent with qualitative observations notably regarding text below STOP signs. Conversely, CLIP-T increases as α decreases, and qualitatively, the generated image tends to be stylized, consistent with qualitative assessments and design intent. DC and CC confirm $\alpha = 0.75$ yields best overall performance. In the local attention transfer module, as shown in Tab. 4, for parameter β , decreasing β can improve the con-

β	DINO↑	CLIP-I↑	CLIP-T↑	DC↑	CC↑
1	0.599	0.627	0.235	1.976	2.010
0.75	0.646	0.630	<u>0.227</u>	2.019	1.999
0.5	0.689	0.650	0.226	2.071	2.023
0.25	<u>0.728</u>	<u>0.682</u>	0.223	2.113	2.057
0	0.739	0.704	0.201	<u>2.088</u>	<u>2.046</u>

Table 4. Quantitative ablation study of hyper-parameter β in Local Attention Transplantation (LAT).

γ	DINO↑	CLIP-I↑	CLIP-T↑	DC↑	CC↑
1	0.746	0.696	0.202	2.099	2.040
0.75	<u>0.733</u>	<u>0.689</u>	0.212	<u>2.101</u>	2.048
0.5	0.728	0.682	0.223	2.113	2.057
0.25	0.714	0.678	<u>0.217</u>	2.086	2.042
0	0.708	0.674	0.212	2.070	2.029

Table 5. Quantitative ablation study of hyper-parameter γ in Style-Infused Noise Initialization (SINI).

tent score (DINO, CLIP-I) while decreasing CLIP-T. Qualitatively, this manifests in the generated image as text and other main subjects gradually becoming blurred, increasing the stylization of the image, consistent with the quantitative results. β weights the query injection, governing content protection versus style transfer, mutually constrained. Based on DC and CC, $\beta = 0.25$ provides optimal balance, validated quantitatively and qualitatively. For γ in the style-infused noise initialization module, as shown in Tab. 5, CLIP-T peaks at $\gamma = 0.5$, while content metrics decline as γ decreases, consistent with enhanced style information at content identity expense. DC and CC indicate $\gamma = 0.5$ offers best stylization outcome, validated quantitatively and qualitatively.

5. Conclusion and Limitations

We propose HAM, a training-free style transfer framework that addresses the core challenge of content-style balance via heterogeneous attention modulation. Our method begins with SINI, followed by two heterogeneous modulation mechanisms: GAR and LAT. Together, these components work synergistically to achieve superior content-style equilibrium. Extensive experiments demonstrate that HAM outperforms the state-of-the-art methods in both fidelity and stylistic quality. Although our method advances content-style balancing, transferring highly abstract or surrealistic artistic styles remains an open challenge for future work.

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